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L_RAG: A Controlled Ablation Study of Vietnamese Legal Question Answering

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Final Project Presentation

Presentation Outline & Primary Focus

1. Introduction
2. System Overview
3. Dataset and Benchmark
4. Ablation Studies & Results
5. Discussion and Conclusion

Core Presentation Focus

This presentation centers on two primary technical pillars:

1. System Overview

5-Stage Pipeline, KG Overlay & Reranking



Empirical Evaluation

2. Ablation Studies

4 Controlled Families (500 Benchmark Cases)

Why Vietnamese Legal QA Is Difficult

Legal requirements

- Answers must be grounded in authoritative statutes.
- Citations must identify the correct article, clause, and item.
- Current validity and amendment history must be respected.
- Many questions require evidence across documents.

Why standard RAG is insufficient

- Dense search may miss exact statutory terminology.
- Fixed chunks break document hierarchy.
- Semantic similarity ignores legal precedence.
- LLMs may hallucinate or cite expired provisions.

Research problem

How can semantic matching, exact lexical search, statutory structure, temporal validity, and reranking be combined in one reliable legal RAG pipeline?

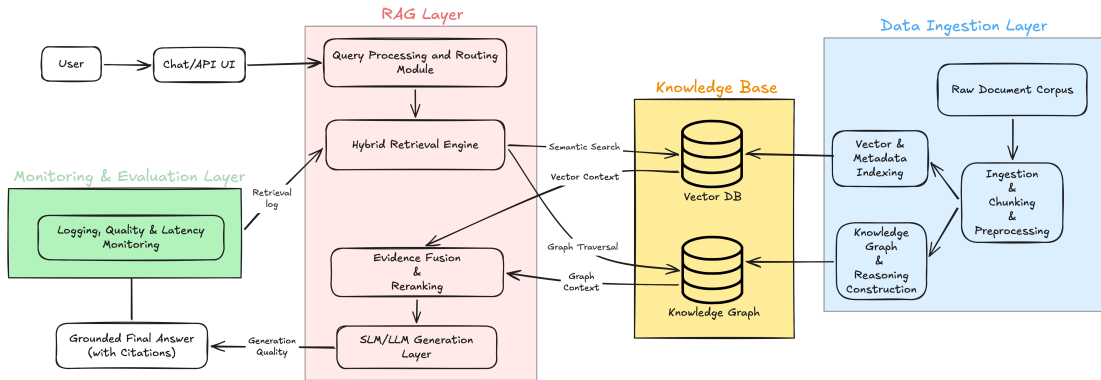
L_RAG: Objectives

- 1 Build an end-to-end hybrid pipeline combining dense E5 retrieval, BM25, and Reciprocal Rank Fusion.
- 2 Construct an in-memory legal knowledge graph for hierarchy, reading order, validity, authority, and context expansion.
- 3 Integrate multilingual rerankers: mMiniLMv2, BGE-Reranker-v2-m3, Jina-Reranker-v2, and Qwen3-Reranker-0.6B.
- 4 Generate answers from selected evidence using `gpt-4o-mini`, with strict citation and abstention rules.
- 5 Construct a standardized benchmark of 500 Vietnamese legal QA pairs.

Target output

A natural-language answer supported by traceable legal chunks, provision-level citations, authority metadata, and validity context.

System Overview: L_RAG at a Glance



End-to-End Visual Architecture

Hybrid Dense-Sparse Search || Knowledge Graph Overlay || Sequential Reranking || Citation-Grounded LLM

L_RAG Pipeline: Stage-by-Stage Processing

Stage 1: Indexing & Representation

- **Dense E5:** Metadata enrichment (Title || Header || Passage) stored in FAISS IndexFlatIP.
- **Sparse BM25:** underthesea tokenizer preserves compound legal terms (trach_nhiem_hinh_su).

Stage 2: KG Overlay Joiner

- **Validity Masking:** Dynamic `id_str_filter` excludes superseded provisions by `as_of_date`.
- **1-Hop Expansion:** Traverses `CHUNK_NEXT` pointers to retrieve sibling context.

Stage 3: Fusion & Sequential Reranking

- **RRF Fusion** ($k = 60$): Merges Dense & BM25 hits into candidate pool C_{30} .
- **Cross-Encoder Reranking:** Sequential GPU load-run-unload cycle yields Top $N = 10$ chunks.

Stage 4: Grounded LLM Generation

- **Model:** gpt-4o-mini ($T = 0.0$, Max tokens 1024).
- **Guardrails:** Strict statutory citation grounding + INSUFFICIENT_CONTEXT abstention.

G-LRAG Knowledge Graph: Schema & Construction

Graph Node Hierarchy

- **Documents (*D*):** Title, Authority rank, Issuing body, Effective date, Validity status.
- **Provisions (*P*):** Article/Clause header, Statutory type.
- **Chunks (*C*):** Passage text, Sequence ID.

Edge Relationships

- **Structural Edges:** DOC_HAS_PROVISION, PROVISION_HAS_CHUNK, CHUNK_NEXT, PROVISION_NEXT.
- **Legal Relations:** Amend, Replace, Cite, Repeal between legal instruments.

Ingested Corpus Scale

Entity Type	Total Count
Documents	151,624
Provisions	1,386,267
Chunks	1,513,376
Document Edges	883,256
External Stubs	19,763

Construction Rule

Draft/unverified records quarantined; dynamic validity whitelisting by `as_of_date`.

Dataset & Benchmark: Evaluation Substrate

Vietnamese legal corpus

Entity	Count
Legal documents	151,624
Hierarchical provisions	1,386,267
Retrieval chunks	1,513,376
Document relationships	883,256
External referenced stubs	19,763

Laws, decrees, decisions, and circulars are normalized into citation-ready chunks retaining document, provision, authority, date, and provenance metadata.

500-query legal QA benchmark

Answerability: 400 answerable + 100 unanswerable

Answer types: abstractive, boolean, extractive, unanswerable

Single-hop	265
Citation	106
Legal validity	96
Multi-hop	21
Cross-document	12

Generated from 3–5-provision bundles, independently verified, diversity-filtered, and deduplicated; verifier pass rate: **72.41%**.

Evaluation contract

Each query stores a reference answer and gold document/provision/chunk IDs. Retrieval uses Recall, Hit, MRR, nDCG, and Precision; generation uses Exact Match, F1/ROUGE-L, answerability, and citation coverage.

Evaluation Metrics: Quality, Evidence & Cost

Primary answer metrics

- **Exact Match (EM):** normalized prediction equals normalized reference: $EM_i = \mathbf{1}[y_i = r_i]$.
- **Token F1:** with multiset overlap O , $P = O/L_y$, $R = O/L_r$, and $F_1 = 2PR/(P + R)$.
- **ROUGE-L:** the same F-score using the token-level longest common subsequence (LCS).
- **Paired delta:** $\Delta_i(m) = m_{\text{CoT},i} - m_{\text{Base},i}$; averaged over shared valid qa_ids with a paired-bootstrap 95% CI.

Scope: lexical metrics use successful answerable outputs; paired statistics use only IDs valid in both runs.

Additional diagnostics

- **Context Recall@k:** $|G \cap R_k|/|G|$, the fraction of gold chunks retrieved.
- **Decision accuracy:** correct answer/refuse decisions divided by scheduled rows (500) or valid rows.
- **Citation presence:** successful answerable answers with a marker divided by successful answerable answers.
- **Structural coverage:** mean share of scored factual sentences carrying a marker.
- **Invalid markers:** invalid markers divided by all markers.
- **Completion:** successful rows / scheduled rows. **Latency:** mean, p50, and p95; milliseconds / 1000 = seconds.

Interpretation boundary

These are automatic overlap, abstention, citation-structure, and efficiency diagnostics—not human judgments of legal correctness, citation entailment, or reasoning quality.

Controlled Experimental Protocol

Shared settings

- Frozen benchmark: 500 QA pairs.
- Retrieval metrics: 400 answerable cases.
- Dense encoder: multilingual-e5-large.
- Non-reasoning runs: gpt-4o-mini.
- LLM ablation: four models, each with Base and CoT.
- GPT-4o-mini pair: fully matched context for causal comparison.
- Temperature 0; maximum 1,024 tokens.
- Comparisons paired by qa_id.

Uncertainty analysis

- Mean per-question deltas.
- Percentile-bootstrap 95% confidence intervals.
- 10,000 resamples for primary comparisons.
- A CI containing zero is treated as inconclusive.

Important

Candidate budgets differ across experiment families, so scores are compared **within a family**, not as one pooled factorial study.

Four Completed Ablation Families

Family	Compared configurations	Candidate/final	Changed factor
Embedding text	Chunk-only; chunk + metadata	10/10	Indexed text
Retrieval/graph	DenseOnly; HybridOnly; DenseGraph; HybridGraph	50/10	BM25 + graph
Reranking	None; RRF; mMiniLM; Qwen3; BGE; Jina	30/10	Final ranker
LLM reasoning	4 models, each with Base and CoT	fixed/10	Prompt mode

All four LLM pairs are reported; GPT-4o-mini has matched contexts, so the other comparisons are descriptive.

Ablation 1: Does Metadata Improve Embeddings?

Variant	Hit@10	Recall@10	MRR@10	Document@10
Chunk-only	53.5%	48.5%	28.6%	65.8%
Chunk + metadata	53.2%	48.5%	22.7%	70.2%
Paired delta	−0.2 pp	+0.1 pp	−5.9 pp	+4.5 pp

Positive effect

Document success improves by **+4.5 pp**; 95% CI [+0.5, +8.8].

Trade-off

Exact-chunk MRR falls by **5.9 pp**; 95% CI [−9.1, −2.7].

Metadata helps locate the correct document, but the first exact gold chunk appears later.

Ablation 2: Dense, Hybrid, and Graph Retrieval

Method	Recall@10	MRR@10	nDCG@10	ID precision	Avg. chunks	Total p50 (s)
DenseOnly	0.4855	0.2267	0.2813	0.0580	10.000	5.483
HybridOnly	0.4257	0.2252	0.2638	0.0493	10.000	31.538
DenseGraph	0.3615	0.2045	0.2374	0.0800	5.605	4.998
HybridGraph	0.3076	0.2012	0.2228	0.0641	6.305	31.180

Additional diagnostics

Method	F1	R-L	Unans. acc.	Total p50/p95 (s)
DenseOnly	0.1949	0.1665	38.0%	5.483/9.946
HybridOnly	0.1990	0.1719	38.0%	31.538/33.623
DenseGraph	0.1987	0.1691	41.0%	4.998/9.750
HybridGraph	0.2014	0.1710	47.0%	31.180/33.312

Hybrid adds latency; graph trades exact-chunk recall for higher identity precision.

Ablation 3: Reranker Comparison

Model	Recall@10	Hit@10	MRR@10	nDCG@10	Token F1	ROUGE-L	Decision	Rerank p50	Post-retrieval p50
None	0.4842	0.5375	0.2167	0.2714	0.2801	0.2453	85.60%	0.000	2.925
RRF-only	0.4382	0.4800	0.2335	0.2728	0.2750	0.2416	87.40%	0.000	2.972
mMiniLM	0.4973	0.5425	0.3047	0.3425	0.2901	0.2568	87.20%	0.385	3.343
Qwen3	0.5218	0.5725	0.3418	0.3688	0.2912	0.2583	87.60%	9.695	13.007
BGE	0.5188	0.5675	0.3247	0.3593	0.2955	0.2603	86.80%	2.872	5.848
Jina	0.5305	0.5850	0.3670	0.3914	0.2913	0.2558	88.40%	1.278	4.158

Best retrieval frontier

Jina gives the best Recall, Hit, MRR, and nDCG with intermediate reranking cost.

Other operating points

BGE slightly leads lexical answer overlap; mMiniLM is the fastest learned reranker.

Key Reranker Insight

Rerankers jointly score each query–candidate pair, capturing richer cross-attention interactions than comparing separately encoded query and document vectors.

Ablation 4: Consolidated LLM Generator Results

Model	Prompt	EM	F1	R-L	Decision	Cite pres.	Struct. cov.	Sources	Gen. p50/p95	Total p50/p95
GPT-4o-mini	Base	14.00	24.72	21.60	82.60%	78.25%	89.42%	1.46	2.92/5.74	13.99/17.21
GPT-4o-mini	CoT	13.75	23.15	19.90	84.00%	81.00%	87.91%	1.68	3.68/6.17	14.74/18.42
DeepSeek-V3.1	Base	22.25	17.84	15.28	92.00%	90.00%	82.38%	2.57	7.28/15.51	16.15/24.55
DeepSeek-V3.1	CoT	22.25	16.51	14.12	92.20%	90.75%	75.21%	2.71	7.82/21.02	18.52/31.37
GLM-5	Base	25.25	16.91	14.90	91.80%	91.75%	80.26%	2.72	4.17/8.79	14.29/19.09
GLM-5	CoT	25.00	16.23	14.27	93.40%	93.75%	78.12%	3.05	4.15/8.73	15.96/21.23
Qwen3-8B	Base	19.25	21.98	18.92	95.80%	97.75%	65.78%	1.79	4.77/9.13	14.15/18.51
Qwen3-8B	CoT	17.25	20.05	17.17	95.20%	98.25%	58.15%	1.81	4.97/10.47	14.58/20.16

Reading boundary

EM, F1, ROUGE-L, and citation fields use 400 answerable outputs; decision accuracy uses all 500. Prompting effects are reported for all four LLMs, but only GPT-4o-mini has fully identical saved contexts.

Additional diagnostics

All eight runs completed 500/500 cases. Invalid markers were zero except for Qwen3 Base (5/877) and CoT (4/901); Context Recall@10 is fixed within each pair and describes evidence availability.

Discussion: Key Findings

- **Metadata:** improves document identification (+4.5 pp) but lowers exact-chunk MRR (-5.9 pp).
- **Hybrid and graph retrieval:** both reduce exact evidence recall under the tested budgets; hybrid adds substantial latency, while graph expansion improves identity precision.
- **Reranking:** every tested cross-encoder improves retrieval quality over RRF-only. Jina leads retrieval metrics, BGE leads lexical overlap, and mMiniLM is fastest.
- **Prompting across models:** across all four LLMs, CoT does not improve EM, F1, or ROUGE-L. It increases latency and lowers structural coverage; decision and citation changes are small and mixed. Only GPT-4o-mini has fully matched contexts, so cross-model comparisons are descriptive.

Scope

Conclusions are limited to the tested corpus, index, budgets, graph policy, and 500-query benchmark.

Limitations and Responsible Use

Validity limits

- 500 questions; difficult slices are small and gold evidence may omit valid alternatives.
- Results depend on one corpus/encoder and unequal candidate budgets.
- Graph validity, semantic correctness, reasoning traces, and cost remain incompletely measured.

Responsible use

- Show citations and validity metadata; abstain when evidence is weak.
- Protect privacy and acknowledge incomplete corpus coverage.
- Require qualified legal review; this is decision support, not legal advice.

Conclusion and Next Steps

Bottom line

The strongest tested configuration is **dense retrieval + cross-encoder reranking**, with **Jina** offering the best quality–latency balance. Metadata helps at document level, but the current hybrid and graph policies need redesign. Across four LLMs, CoT adds cost without improving lexical answer quality.

Next steps

- Add blinded labels for correctness, faithfulness, completeness, and citation quality.
- Tune candidate pools, RRF, graph budgets, and seed-preserving expansion.
- Align provenance and rerun incomplete model comparisons.
- Test broader Vietnamese benchmarks; measure end-to-end p95, memory, throughput, and cost.

The results are benchmark-bounded engineering evidence—not certification of legal reliability.

Questions & Discussion

Thank you!